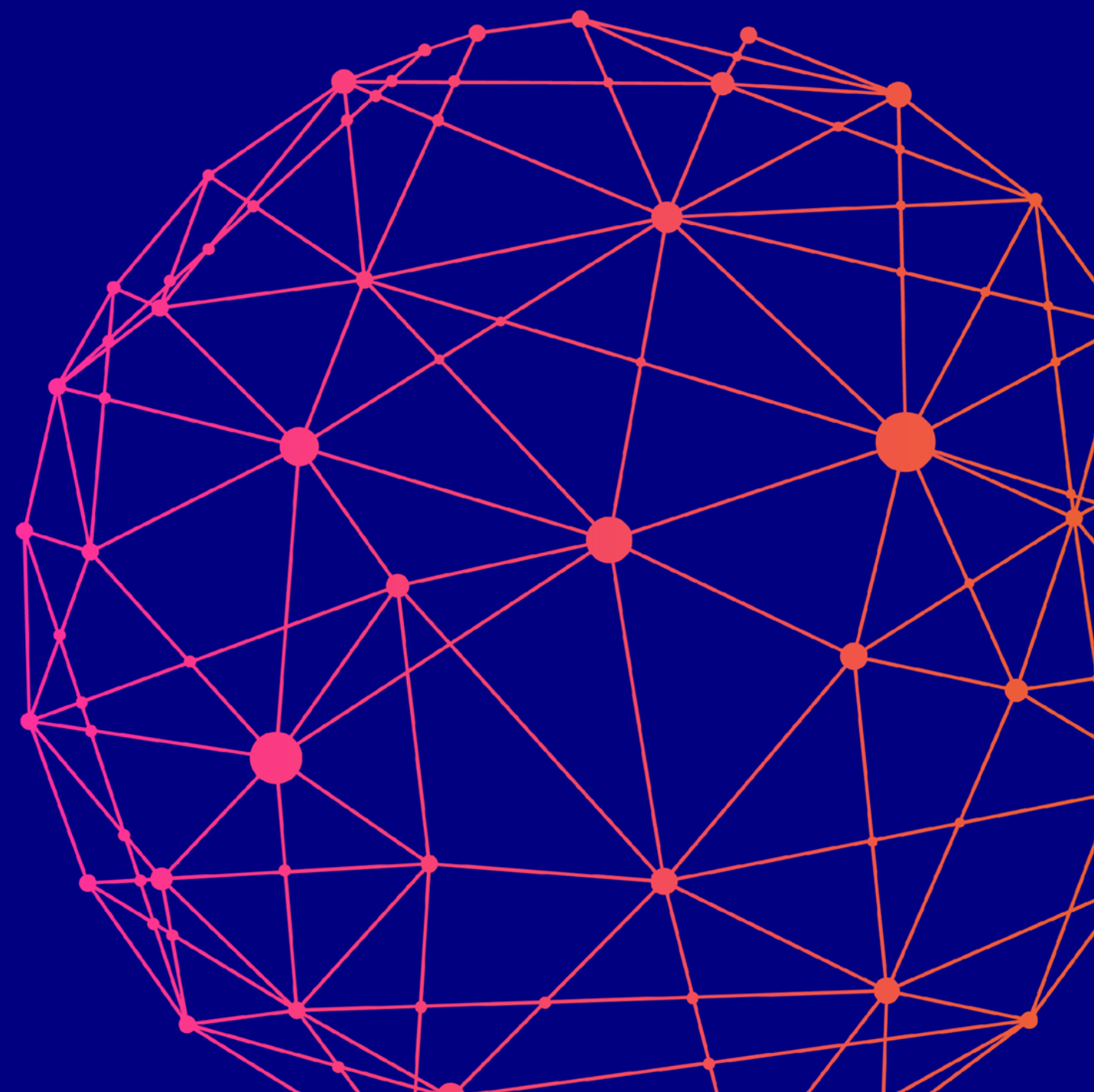


Data Spaces Symposium

Accelerating adoption. Increasing impact.

Scientific data
for European
AI in science

David ARRANZ (X: @David_Arranz_)
Policy Officer – AI in Science
DG RTD.E4 European Commission



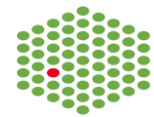
Scientific data behind major AI and scientific breakthroughs

High-quality scientific datasets → more robust models → discoveries and breakthroughs



Protein Data Bank (PDB)

wwPDB consortium / RCSB PDB



AlphaFold

Revealing millions of intricate 3D protein structures, and helping scientists understand how life's molecules interact.

Google DeepMind



Breakthrough: Protein structure prediction at scale



ERA5

ECMWF (multi-decade reanalysis)



WeatherNext 2

Our most accurate AI weather forecasting technology

Google DeepMind

Breakthrough: Global weather forecasts and extreme events



Drug Repurposing Hub

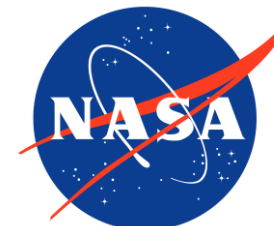
Broad Institute compound library



Halicin screening

MIT (Collins Lab et al.)

Breakthrough: AI-discovered antibiotic candidates



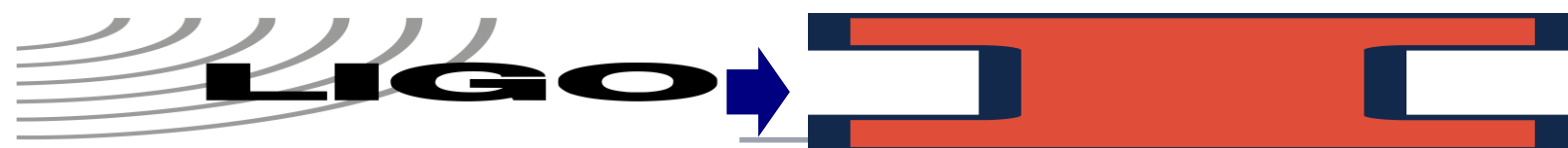
Kepler light curves

NASA Kepler mission data



Google AI (+ collaborators)

Breakthrough: Exoplanet discovery via deep learning



LOSC strain/noise

LIGO Open Science Center



Deep Filtering

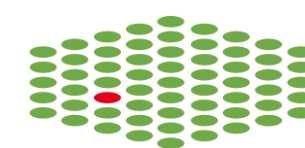
UIUC / NCSA (+ collaborators)

Breakthrough: Real-time GW detection & parameter estimation

EMBL

EMPIAR cryo-EM

EMBL-EBI archive of raw cryo-EM data



cryoDRGN

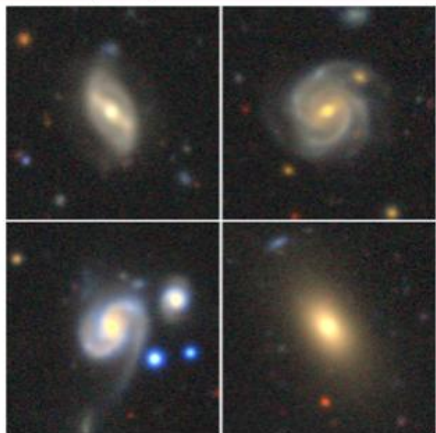
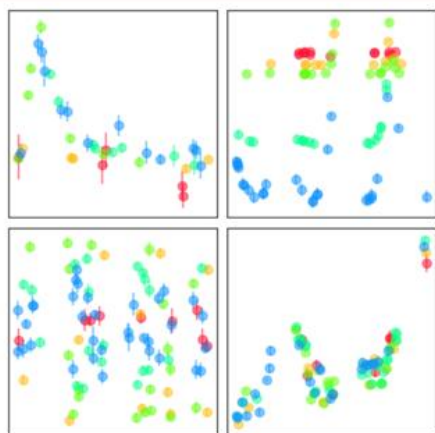
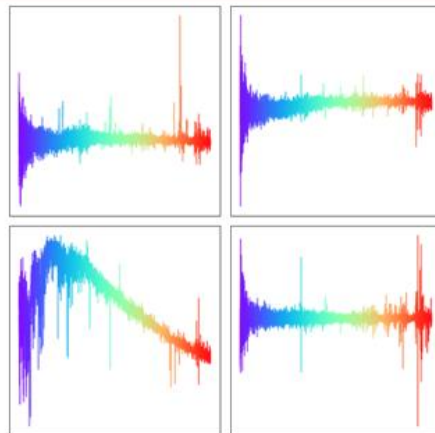
MIT (+ collaborators)

Breakthrough: Modeling structural heterogeneity with generative AI

More scientific datasets for new challenges

The Multimodal Universe

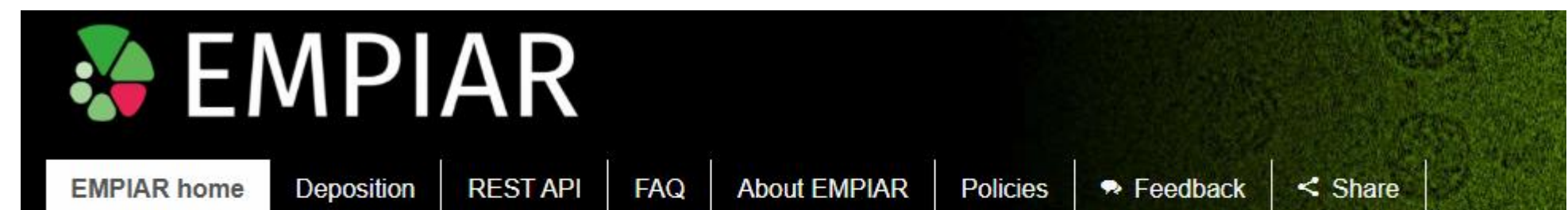
100TB of Astronomical
Scientific Data

	Images	Time-Series	Spectra
# examples	140M	4.5M	225M
Description	images in a variety of wavelength ranges, including optical and infrared	multivariate time-series of flux + uncertainty in different wavelength ranges	flux as a function of wavelength
Tasks	galaxy classification, physical property estimation	time-series classification, redshift estimation	physical property estimation
Examples			



TheWell

Bridging the Gap Between
Physical Numerical Simulations
and Machine Learning.
15 datasets - 15TB



EMPIAR, the Electron Microscopy Public Image Archive, is a public resource for raw images underpinning 3D cryo-EM maps and tomograms (themselves archived in [EMDB](#)). EMPIAR also accommodates 3D datasets obtained with volume EM techniques and soft and hard X-ray tomography. [More ...](#)

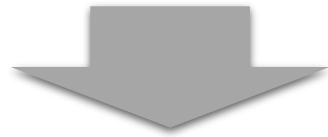
As of **2026-01-30**, EMPIAR contains **2678** entries, taking up **7.73 PiB** of storage.

What do we need and what can policy do?

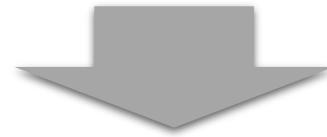


EU AI Framework

AI Continent Action Plan



Apply AI Strategy



AI in Science
Strategy



Data Union Strategy

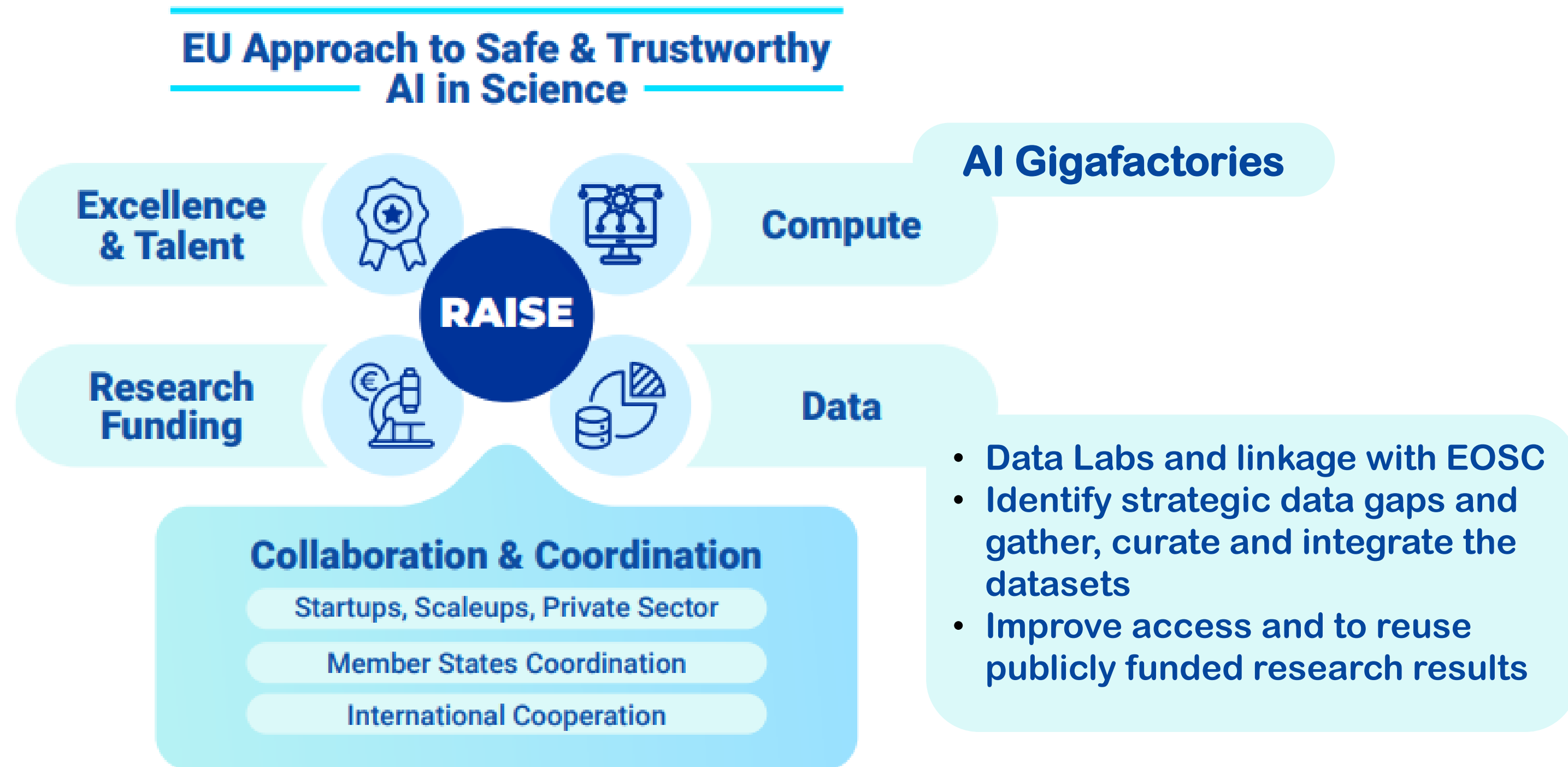
Cloud and AI
Development Act

AI Gigafactories

...

RAISE
(Resource for AI Science in Europe)

AI in Science Strategy & RAISE



Other strategies to scientific Data and AI



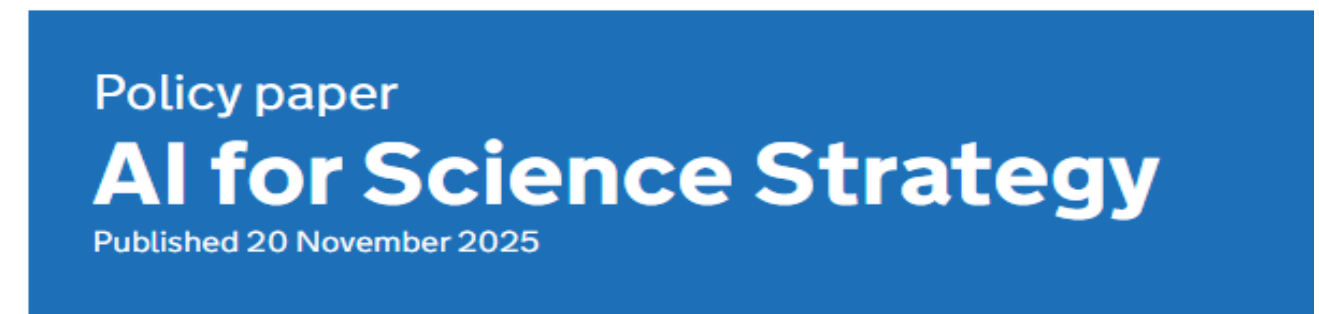
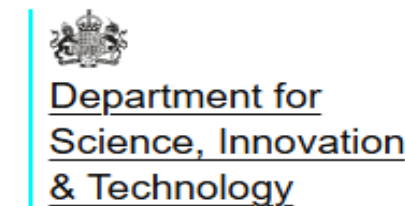
Central idea: an integrated AI platform built around Federal scientific datasets

integrated AI platform designed to harness Federal scientific datasets (described as the world's largest such collection) to train scientific foundation models and power AI agents for hypothesis-testing and workflow automation.

Data readiness

Within 120 days, DOE must identify initial data/model assets for the Mission

Interagency data alignment and interoperability



Pillar 1: Data

quality matters, datasets are only constitutive parts of a larger whole what's good today might not be good tomorrow

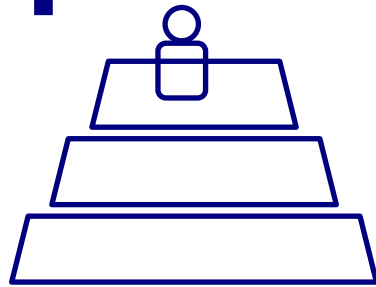
Our objective is not to achieve maximum data generation, but to ask: **'what data do we need to solve the challenges we face?'**

Priority on new data: Newly generated datasets have the advantage of being precisely tailored to specific research challenges.

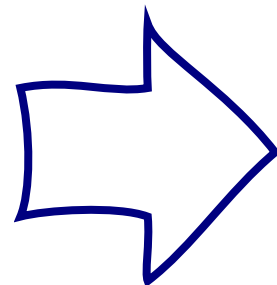


Policy Mix

Top-down



proactive

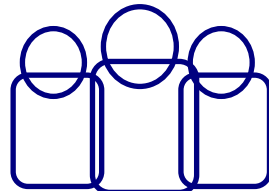


**Scientific Data
for
Scientific AI**



demand

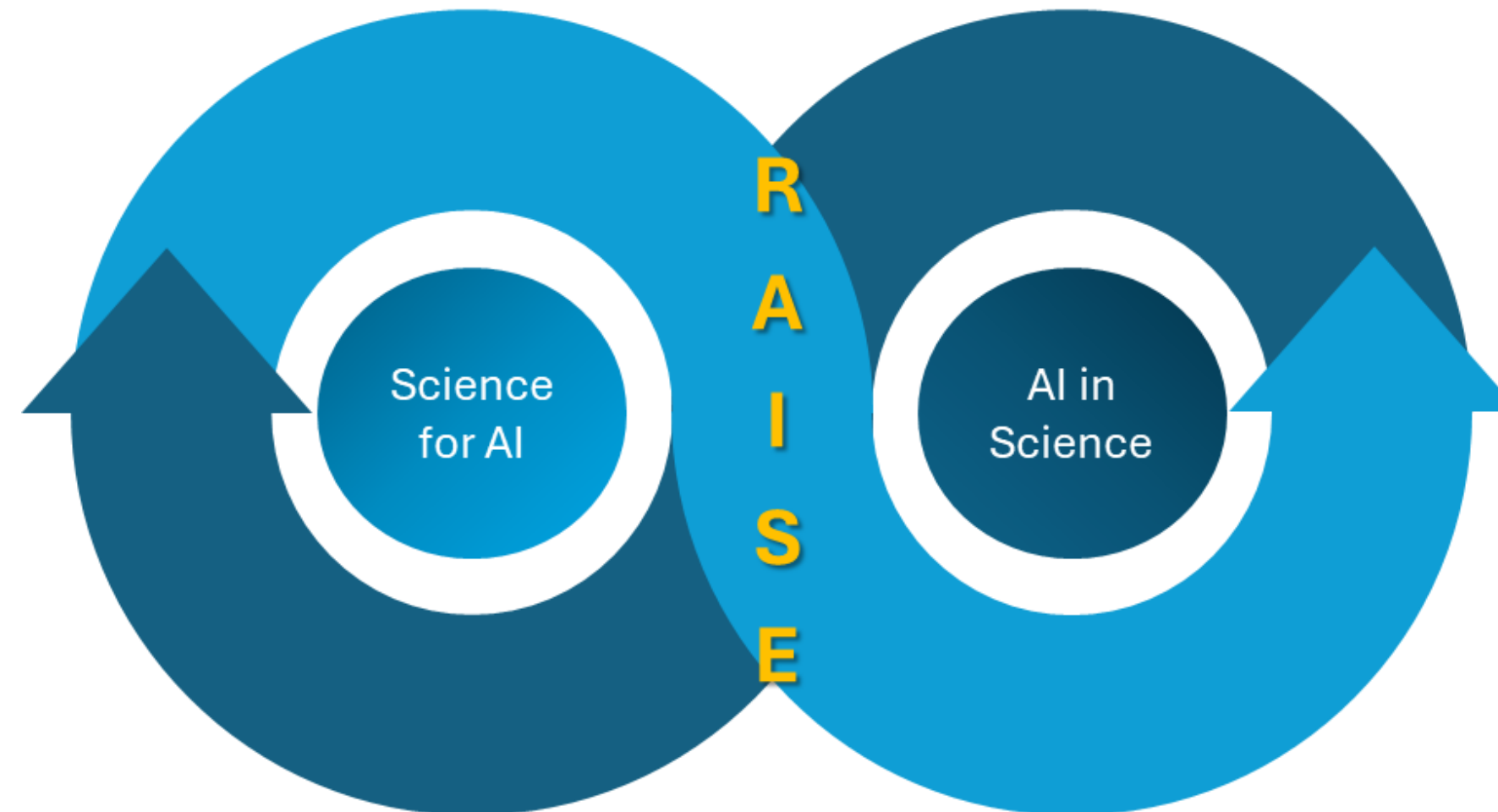
Bottom-up



RAISE

The Resource for AI Science in Europe

What is it? Virtual European institute that pools, aligns and coordinates essential AI resources. Promoting cutting-edge research in AI and the uptake of AI for scientific progress across disciplines (AI in science).



What does it offer? A dynamic research community to foster interdisciplinary collaborations between the best scientists in Europe. AI resources: long-term funding, EU's computational power and datasets at a scale.

How does it work? Interconnection and collaboration between science for AI and AI in science, Member States and the private sector. A high-level academic advisory board provide scientific guidance. A secretariat will ensure the symbiosis of RAISE elements

The case for materials science

